What can a robot remember about a cube?

A simulated Panda arm interacts with cubes whose mass and friction can vary. Two experiments examine different uses of physical information.

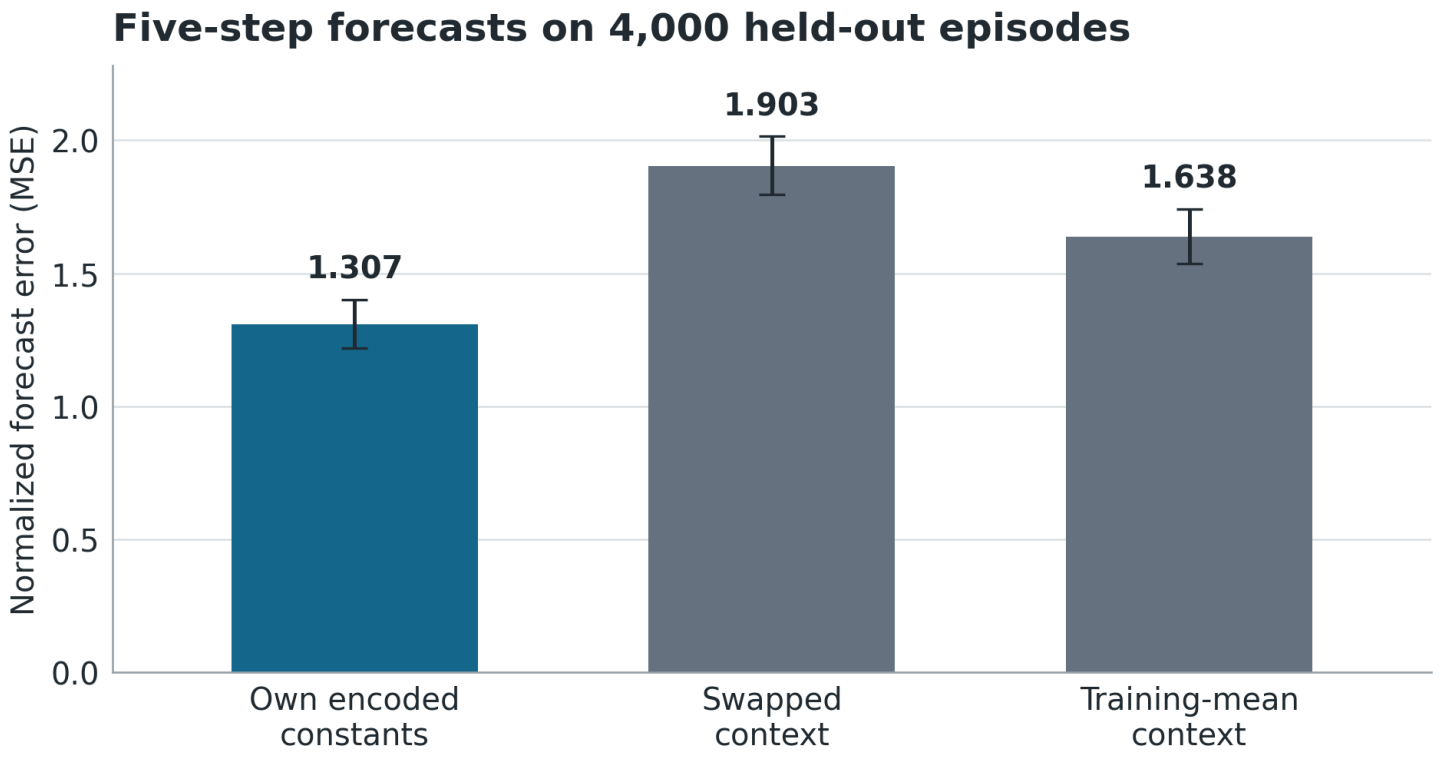
Forecasting motion

How does supplied physics change a forecast?

Remembering a mass estimate

What carries over to the next try with the same cube?

Supplied physics improved motion forecasts



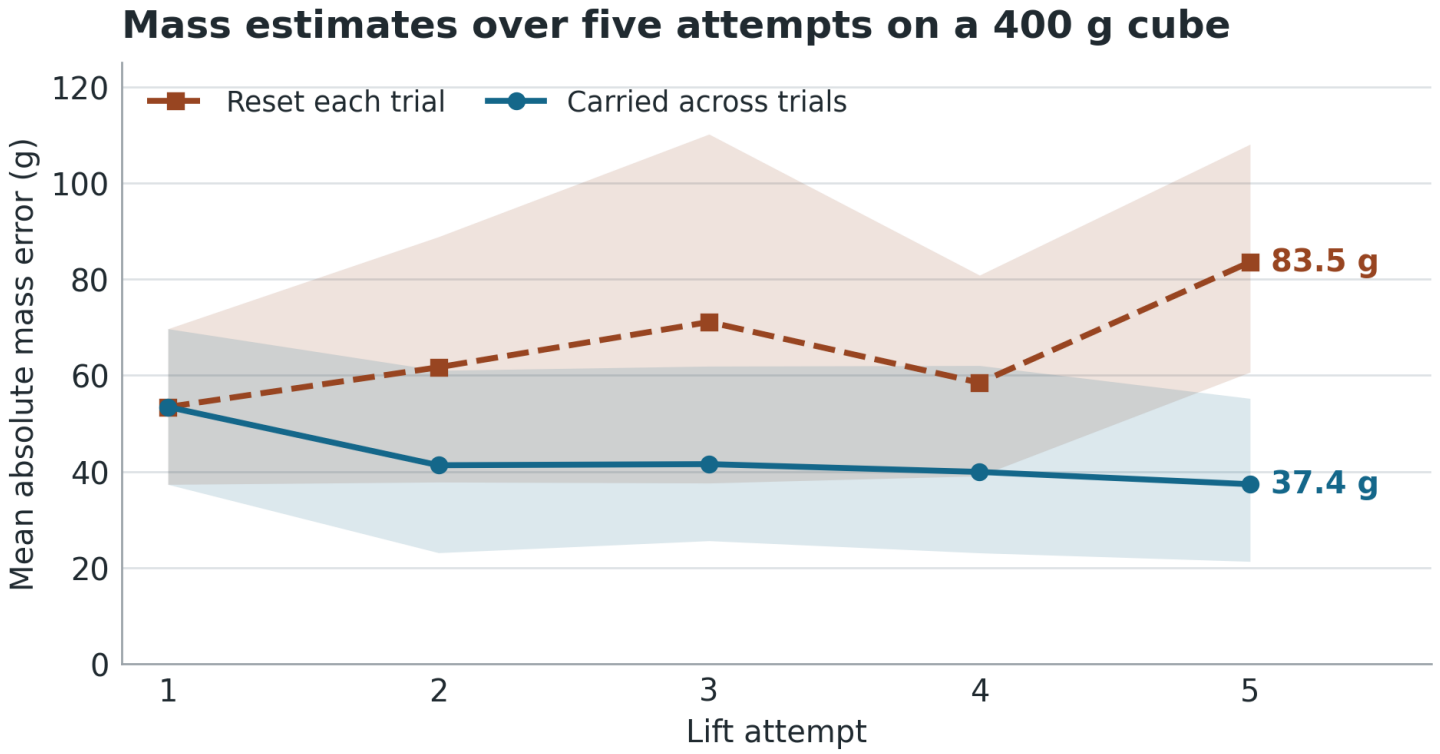
MSE is mean squared prediction error. Whiskers show 95% intervals across whole episodes.

45.6%

higher forecast error after swapping the context

The context encoded ten true simulator settings for the cube, robot and table. The score averages all steps, including free motion.

Reusing the estimate reduced fifth-trial mass error



Bands show 95% intervals from resampling 12 complete sequences of five attempts each.

46.2 g

less error on trial five with the carried estimate

Paired 95% interval 14.6–73.7 g. Both methods saw the same trials. A frozen estimator used recent commands, joint motion and finger compression.

Will experience with earlier cubes improve the first guess for a new cube?

What to vary	Draw new cubes from a mostly heavy population. Give each cube a stable identity and record both pushes and lifts.
What to compare	Use earlier cubes' inferred estimates to set a first guess. Compare it with a fixed first guess on identical observations of each new cube.
What to measure	Score initial error and steps to a prespecified accuracy threshold. Retain cubes that never reach it and report their count.